

Exploring Graphs and Networks with Worked Examples

NetworkX Explorer · NetworkX

Senior Secondary (Years 11-12)

~30 minutes

Mathematics · General
Mathematics (Networks)

NOTICE — AI-generated worksheet

This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://networkx.org/documentation/stable/auto_examples/index.html

Curriculum links: graph theory · networks

Learning intentions

- I can identify vertices, edges and paths in a network diagram.
- I can describe how a shortest-path example finds an efficient route through a network.
- I can compare different network examples and describe features of their structure, such as degree and connectivity.

Getting started

Open the NetworkX examples gallery and browse the shortest-path and network-structure sections. Choose one shortest-path example to examine closely, looking at both the diagram and the accompanying code to see how the network is built.

Tasks

1. Choose one network diagram from the gallery. Count and record the number of vertices (nodes) and edges shown.

2. For your chosen network, identify a vertex with a high degree (many connections). Explain what a 'high degree' vertex might represent in a real network, such as a hub airport or a well-connected person.

3. Find a shortest-path example in the gallery. Explain, in your own words, what 'shortest path' means in this context — is it always about physical distance?

4. Trace the shortest path shown between two labelled vertices in your chosen example. List the vertices the path passes through, in order.

5. Choose three different network diagrams from the gallery. Record the number of vertices and edges in each.

Example name	Vertices	Edges

6. Compare two of the networks in your table. Which one looks more 'connected' or dense, and what evidence from the diagram supports this?

7. Explain one real-world situation, such as road networks, social networks or delivery routes, where finding the shortest path through a network would be genuinely useful, and describe what the vertices and edges would represent.

Extension (optional)

A minimum spanning tree connects every vertex in a network using the least total edge weight, without any cycles. Browse the gallery for an example that could represent this idea, and explain how it differs from a shortest-path example between just two vertices.
